

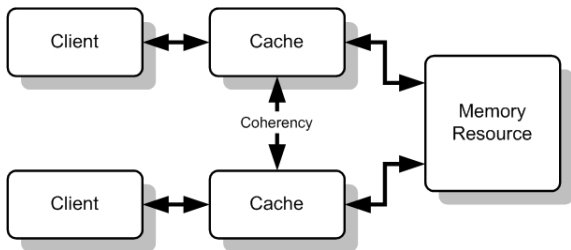
Lecture 8 — Cache Coherency

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Each CPU has its own cache: coordination is needed!

This applies to CPU caching but also things like distributed redis/redisct.

Coherency:

- Values in all caches are consistent;
- System behaves as if all CPUs are using shared memory.

Initially in main memory: $x = 7$.

- 1 CPU1 reads x , puts the value in its cache.
- 2 CPU3 reads x , puts the value in its cache.
- 3 CPU3 modifies $x := 42$
- 4 CPU1 reads $x \dots$ from its cache?
- 5 CPU2 reads x . Which value does it get?

Unless we do something, CPU1 is going to read invalid data.

Planes and Trains



The simplest way to “do something” is to use Snoopy caches.

No, not this kind of Snoopy (sadly):



High-Level Explanation of Snoopy Caches

Snoopy: the caches are spying on each other.

- Each CPU is connected to a simple bus.
- Each CPU “snoops” to observe if a memory location is read or written by another CPU.
- We need a cache controller for every CPU.

What happens?

- Each CPU reads the bus to see if any memory operation is relevant. If it is, the controller takes appropriate action.

This is a distributed approach; no centralized state is maintained.

Each cache with a copy of data from a block of physical memory knows whether it is shared or not.

Whenever a CPU issues a memory write, the other CPUs are watching (snooping around) to observe if that memory location is in their cache.

If it is, then the CPU will need to take action.

What does action mean?

The Air Canada action was **update**.

The Deutsche Bahn action was **invalidate**.

Simplest type of cache coherence:

- All cache writes are done to main memory.
- All cache writes also appear on the bus.
- If another CPU snoops and sees it has the same location in its cache, it will either *invalidate* or *update* the data.

Our chosen approach:



Invalidation is the most common protocol.

It means the data in the cache of other CPUs is not updated, it's just noted as being out of date (invalid).

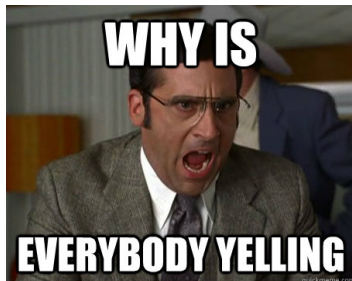
If we want to do a read and there's a miss, we can poke around in other caches to see who has the most recent cached version.

This is a bit like going into a room and yelling “Does anybody have block...?”, in some sort of multicast version of the card game “Go Fish”.

Regardless, the most recent value appears in memory, always.

There are also write broadcast protocols, in which case all versions in all caches get updated when there is a write to a shared block.

But it uses lots of bandwidth and is not necessarily a good idea.



It does, however, prevent the costly cache miss that follows an invalidate.

Sadly, as we are mere users and not hardware architects, we don't get to decide; we just have to live with whichever one is on the hardware we get. Bummer.

- Two states, **valid** and **invalid**, for each memory location.
- Events are either from a processor (**Pr**) or the **Bus**.

This table is for write-through write-allocate:

State	Observed	Generated	Next State
Valid	PrRd		Valid
Valid	PrWr	BusWr	Valid
Valid	BusWr		Invalid
Invalid	PrWr	BusWr	Valid
Invalid	PrRd	BusRd	Valid

There are write-allocate and write-no-allocate variants of write-through. In all cases, the written value goes to memory.

- write-allocate: Invalid/PrWr line has next state Valid: the written value is also immediately cached.
- write-no-allocate: Invalid/PrWr line has next state Invalid: the written value is not immediately cached.

Write-Through Protocol Example

- For simplicity (this isn't an architecture course), assume all cache reads/writes are atomic.

Using the same example as before:

Initially in main memory: $x = 7$.

- 1 CPU1 reads x , puts the value in its cache. (valid)
- 2 CPU3 reads x , puts the value in its cache. (valid)
- 3 CPU3 modifies $x := 42$. (write to memory)
 - CPU1 snoops and marks data as invalid.
- 4 CPU1 reads x , from main memory. (valid)
- 5 CPU2 reads x , from main memory. (valid)

- What if, in our example, CPU3 writes to x 3 times?
- Main goal: delay the write to memory as long as possible.
- At minimum, we have to add a “dirty” bit:
Indicates our data has not yet been written to memory.

The simplest type of write-back protocol (MSI), with 3 states:

- **Modified** – only this cache has a valid copy; main memory is **out-of-date**.
- **Shared** – location is unmodified, up-to-date with main memory; may be present in other caches (also up-to-date).
- **Invalid** – same as before.

Initial state, upon first read, is “shared”.

Implementation will only write the data to memory if another processor requests it.

During write-back, a processor may read the data from the bus.

- Bus write-back (or flush) is **BusWB**.
- Exclusive read on the bus is **BusRdX**.

State	Observed	Generated	Next State
Modified	PrRd		Modified
Modified	PrWr		Modified
Modified	BusRd	BusWB	Shared
Modified	BusRdX	BusWB	Invalid
Shared	PrRd		Shared
Shared	BusRd		Shared
Shared	BusRdX		Invalid
Shared	PrWr	BusRdX	Modified
Invalid	PrRd	BusRd	Shared
Invalid	PrWr	BusRdX	Modified

Using the same example as before:

Initially in main memory: $x = 7$.

- 1 CPU1 reads x from memory. (BusRd, shared)
- 2 CPU3 reads x from memory. (BusRd, shared)
- 3 CPU3 modifies $x := 42$
 - Generates a BusRdX.
 - CPU1 snoops and invalidates x .
 - CPU3 changes x 's state to modified.
- 4 CPU1 reads x :
 - Generates a BusRd.
 - CPU3 writes back the data and sets x to shared.
 - CPU1 reads the new value from the bus as shared.
- 5 CPU2 reads x from memory. (BusRd, shared)

The most common protocol for cache coherence is MESI.

Adds another state:

- **Modified** – only this cache has a valid copy; main memory is **out-of-date**.
- **Exclusive** – only this cache has a valid copy; main memory is **up-to-date**.
- **Shared** – same as before.
- **Invalid** – same as before.

MESI allows a processor to modify data exclusive to it, without having to communicate with the bus.

MESI is **safe**: in E state, no other processor has the data.

MESIF (used in the latest i7 processors):

- **Forward** – basically a shared state; but, current cache is the only one that will respond to a request to transfer the data.

Hence: a processor requesting data that is already shared or exclusive will only get one response transferring the data.

Permits more efficient usage of the bus.



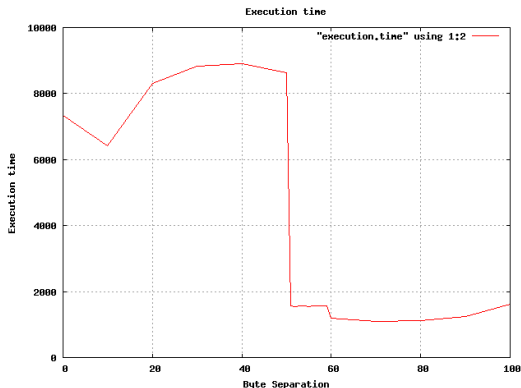
False sharing: program has two unrelated data elements that are mapped to the same cache line/location.

```
char a[10];  
char b[10];
```

They don't overlap but are located next to each other in memory.

Solution: Heap allocation? Maybe...

Make both arrays bigger, wasting some space...?



Is wasting a little space worth it? Yes!

(Plus, putting arrays in a struct and padding enables future updates to the struct.)

Write a pseudocode description of behaviour for a distributed software cache that uses the MESI states and has write-back behaviour.

This cache is for data items retrieved from a database, so if the item is not in any node's cache, write down retrieve item i from database.

Assume the cache to be huge – no need to worry about replacement.

As a practice problem for consideration, think about what modification(s) you would need to make for that scenario.

What happens if a node comes online and joins the cluster, and what happens if a node is going to shut down and leave the cluster.

You may ignore situations like crashes or network outages and you can assume all sent messages are reliably delivered/received.

```
Add to cache ( item i ) { // Helper function
  if cache is not full {
    add i to cache
    return
  } else {
    old = least-recently-used item
    if old state is M {
      write old to database
    }
    replace old with i
  }
}
```

```
Current Node Startup {
  get known_nodes from coordinator
  Send message to known_nodes to indicate joining
}
```

```
New Node Joins ( node n ) {
  Add n to known_nodes
}
```

Let's fill these in (whiteboard)

Current Node Shutdown { }

Node Leaves (node n) { }

Get Item (item i) { }

Other Node Searching (item i) { }

Update Item (item i) { }

Invalidate (item i) { }

Solutions in the notes!

We saw the basics of cache coherence (more of an architecture thing).

Knowing this stuff allows us to make wise decisions (e.g., false sharing)!

There are other protocols for cache coherence, each with their own trade-offs.